

Response to reviewer and editor comments

Manuscript: “A real-time reservoir inflow forecast evaluation framework” (MS 8299185), submitted to the *Journal of the American Water Resources Association*.

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We thank the editor and the four reviewers for their comments. The reviewers and the handling editor converged on one point: the manuscript described a framework without stating what was new about it. We agree. The original draft presented the five-step process as though the process itself were the contribution. We have added an explicit statement of contribution to the introduction, restructured the process description so that the generic steps are labeled as such, added a discussion subsection comparing the framework against retrospective verification.

We have also expanded the description of the study system and the forecast products, added a smoothing sensitivity analysis, characterized the hydrology of the evaluation period, cited the forecast value literature, and corrected several errors in the methods.

Four statements in the original manuscript have been corrected:

- The evaluation period was given as July 2023 to May 2025. The record analyzed runs from June 2023 to April 2025.
- The three forecast products do not cover the period uniformly. Forecast B spans the full period; forecast A and the in-house forecast both end in November 2024. The metrics in Table 1 are therefore computed over overlapping but not identical sample sizes. We now report the sample size for each product and have verified that restricting the comparison to the 510 days on which all three are present does not change the ranking at any location.
- The paragraph comparing metrics across the three dams labeled the reported values as KGE when they were NSE. This is the passage Reviewer 1 asked us to check (comment 1.6), and rewriting it is the largest change to the results.
- The monthly KGE values cited for forecast A in January and February were 0.39 and -0.40 . Recomputing gives 0.45 and -0.67 .

Reviewer comments are set in indented italics, trimmed to the operative text where they are long. A tracked-changes manuscript accompanies this response, so we quote revised text only where the wording is itself the answer.

Editor comments

The greatest barrier [...] is the question of novelty and reproducibility. Reviewers 1, 3, and 4 note that there is no distinct statement of novelty, with reviewer 3 particularly noting that the manuscript seems to suggest that the framework is the novelty without fully describing the novelty.

Agreed. We have added a statement of contribution to the end of the introduction:

“The contribution of this work is a specific operating procedure for the case where a utility must choose between competing forecast products under live operating conditions and cannot run a hindcast. That procedure has two components that distinguish it from retrospective verification practice. First, the operators participate in metric selection and design and can

modify, remove, or append metrics to the evaluation as more data is collected. Second, the evaluation runs on the same operational data the operators use which reflects the real day-to-day operational decisions.”

We have also rewritten the preceding paragraph to be more clear.

“Traditional verification requires hindcasts over a climatologically representative period, which is expensive and may be infeasible. The fallback of evaluating archived operational forecasts fits poorly with products that are updated continuously, trained on evolving data streams, hand adjusted by operators, or developed by third parties.”

On reproducibility, we have anonymized the proprietary data and made the forecast evaluation as reproducible as we could given the limitations of the data. The evaluation code is public (<https://github.com/HydroWIRES-PNNL/inflow-forecast-evaluation>) and sample data are on Zenodo (<https://doi.org/10.5281/zenodo.16921728>). The repository now also includes a script that reproduces every number reported in the paper from the published data. We have added the parameter values that were previously left implicit, most importantly the smoothing window (comment 4.1), together with a sensitivity analysis. The proprietary forecast time series cannot be released, which we state as a limitation.

Reviewer 1

The topic and the contents are relevant, and it is very interesting for hydropower operators and decision-makers. However, the novelty and main contributions of this study are unclear.

Thank you for the review. The comments on framing, the missing system description, and the table and figure quality were all warranted. Comment 1.6, asking us to check an apparent anomaly in the NSE values, uncovered a mislabeling in that paragraph and has become a result in the revised manuscript.

1.1 Novelty

The novelty and main contributions of this study are unclear. Please explain and discuss in detail.

Please see our response to the editor for the new statement of contribution. In brief, the contribution is a procedure for the case where a utility must choose among competing forecast products under live operating conditions and a hindcast is not possible. What distinguishes it from standard forecast verification is the direct operator input in the process, and that the evaluation runs on operational data rather than an archived dataset.

We have added to the discussion why this matters now. Forecast products using AI and machine learning methods are being marketed to utilities at an increasing rate, and the verification statistics vendors supply may not be verifiable. A utility generally has no standard procedure for checking those claims against its own system, and as we now report, often cannot inspect the model at all.

1.2 Persistence as a benchmark at long lead times

[...] Fig. 3 and Fig. 4 show that 10-day and months “persistence forecasts” as benchmark which is unrealistic in practice.

You are correct, persistence is often hard to beat at short lead times. We had stated that persistence is competitive to about two days and then plotted it out to ten. Reviewer 2 raised a related point in comment 2.7. We have revised the benchmark section:

“Persistence is a simple but competitive forecast at short lead times, from hours to roughly two days, and is the appropriate reference for day-ahead operations, which is the horizon of interest for the system evaluated here. It degrades quickly beyond that horizon, where a climatological forecast is the more appropriate lower bound. Evaluators should match the benchmark forecast to the horizon of interest: persistence for hourly to 2-day operations, climatology for weekly to seasonal horizons. Where persistence is plotted at longer lead times in the figures below, it is retained only to show the scale of degradation and should not be read as a meaningful reference at those horizons.”

The figure captions now carry the same qualification.

We have not added a climatological benchmark to the figures. A defensible daily climatology for these locations requires a long record of calculated inflow at the dams. While there are nearby USGS gages, these reflect older operations and so would not give a representative climatology.

1.3 Operating problems and the forecasting scheme in practice

What are the main operation problems of these hydropower stations? Which forecasting scheme is used in practice?

We have added a subsection, “Operating context.” The three dams have limited storage relative to their inflow and are operated as run-of-river facilities, with a daily cycle of an upward ramp, flexible generation, a downward ramp, and refill. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most. The binding constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation with penalties for exceedance. The central operating problem is committing generation a day ahead on an uncertain inflow forecast without violating the forebay limits, which is what motivated the cumulative-error and threshold-exceedance metrics.

On the scheme used in practice: the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent against which the two commercial forecasts were evaluated, labeled GRH in the results, and it extends only about 32 hours because the day-ahead bid is the only decision it is used for.

1.4 Description of the forecast products

Two commercial forecasts (A, B) and the GRH in-house forecast (GRH) should be described in detail, such as what type of hydrologic model used? How about the calibration and validation results, what type input data requirement etc.

Reviewers 2 and 4 raised the same point (comments 2.16 and 4.3). We agree that anonymization without any methodological characterization leaves readers unable to interpret the results. We have added a subsection describing what we are able to disclose:

“Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the Connecticut River mainstem, driven by a commercial numerical weather prediction forecast, and issued once daily at an hourly timestep out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at hourly timestep out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and

validation diagnostics, and we did not have access to them. A utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework.”

The issue frequencies and horizons are taken from the forecast archive rather than vendor documentation, so they describe what was actually delivered.

We cannot provide calibration and validation results. The data used in this study was provided on the basis that it be anonymized. The lack of detailed information about the models reflects the reality that utilities face when procuring commercial forecasts. We have also added that a full procurement evaluation should weigh ongoing operational costs, data dependencies, and ease of integration alongside skill, since these may differ substantially between commercial products.

1.5 Watershed information

[...] *The watershed information, such as area, river length etc. should be described in detail.*

Agreed. We have added a table of physical characteristics for all six evaluation points, with drainage areas from the USGS for the reference gage at each location:

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottauquechee River	Tributary	01151500 North Hartland, VT	221	572

We have also added text describing the cascading system. The mainstem areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder. Because releases from each dam become inflow to the next, an error in an upstream forecast propagates to the downstream forecast and compounds with lead time, which we now quantify: for forecast A, day-ahead PBIAS at Bellows Falls is 0.4% and grows to −19.1% by day 7.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers.

1.6 NSE values in Table 1

In Table 1, the NSE values of forecast scheme A at Wilder, Bellows and Vernon watersheds are 0.72, 0.55, 0.81, respectively. Please check the evaluation results. The forecasting accuracy at the Bellows might high than that at the Wilder.

Thank you for checking this; it was the most productive comment we received. The three NSE values are correct as published, but following up on why the pattern looks wrong uncovered both an error in the text and a result we had missed.

The error first: the paragraph discussing these values described them as KGE when they are NSE, and then drew a conclusion about KGE from them. That paragraph has been rewritten.

On the substance, you were right that the cross-site pattern should not be taken at face value, though for a different reason than the one you proposed. NSE is normalized by the variance of the observations at each location, so NSE values are not comparable across locations with different

inflow variability (Knoben et al., 2019). Flows become progressively more regulated and therefore smoother downstream, and smoother flows yield higher NSE for the same quality of forecast. The persistence benchmark makes this visible: persistence alone achieves NSE of -0.81 at Wilder, 0.39 at Bellows Falls, and 0.80 at Vernon. At Vernon, carrying yesterday’s inflow forward already explains most of the variance, so little room is left for a forecast to improve on it.

The fix is to change what the metric is referenced to rather than to change the metric. NSE is already a skill score, measured against the mean of the observations, and that reference is easy to beat at Wilder and hard to beat at Vernon, which is the whole of the discrepancy you noticed. We have therefore added an NSE skill score against persistence as a column of Table 1, defined by a new equation:

$$\text{NSE}_{SS} = \frac{\text{NSE}_f - \text{NSE}_{ref}}{1 - \text{NSE}_{ref}} = 1 - \frac{\text{MSE}_f}{\text{MSE}_{ref}}$$

The second equality is the useful one: the score is the fraction of the reference forecast’s squared error that the forecast under evaluation removes. So that the column can be checked directly, it is recoverable from the RMSE column of Table 1 as $1 - (\text{RMSE}_f/\text{RMSE}_{ref})^2$. For forecast A:

Location	NSE	Persistence NSE	NSE _{SS} vs persistence
Wilder	0.72	-0.81	0.85
Bellows Falls	0.55	0.39	0.25
Vernon	0.81	0.80	0.03

The ordering inverts. Forecast A is strongest at Wilder, not weakest, and at Vernon it scores 0.03, reducing the squared error of a persistence forecast by three percent. The site where its raw NSE is highest is the site where it adds almost nothing to what an operator would get by assuming today’s inflow will repeat; restricting to the 510 days common to all three products puts it slightly below persistence there (-0.01). The other two products retain real skill everywhere: 0.91 and 0.94 at Wilder, 0.85 and 0.77 at Bellows Falls, and 0.44 and 0.47 at Vernon for forecast B and the in-house forecast.

So the numbers are right, the cross-site pattern in raw NSE is a normalization artifact rather than a ranking of forecast quality, and the pattern that survives correction is not the one the raw values suggest. On your specific suggestion that accuracy at Bellows Falls may be higher than at Wilder, the skill score says the opposite, 0.85 against 0.25 , and the new RMSE column agrees in absolute terms, $2,278$ cfs against $5,904$ cfs. It is Vernon rather than Bellows Falls that the raw values misrepresent.

The methods now recommend reporting skill against a deliberately chosen benchmark alongside raw metrics in any multi-site evaluation, and the discussion carries the general lesson: a raw efficiency metric tells you a forecast explains most of the variance, while skill against a benchmark tells you whether it improves on the alternative the operator already has.

1.7 Table format

The format of Tables in the manuscript are poorly designed and needed be changed.

Agreed. Table 1 has been rebuilt with rows grouped by location, the location named once per group, numeric columns aligned, consistent significant figures, and units in the column headers. It has gained three columns: the number of day-ahead forecasts evaluated for each product, which is necessary because the products cover different portions of the period; RMSE in cfs (comment 4.7); and the NSE skill score against persistence (comment 1.6). Three tables have been added, giving basin characteristics (1.5), the smoothing sensitivity analysis (4.1), and the provenance of

each tailored metric (3.2).

1.8 Figures

The figures also need be redrawn.

We have revised the captions substantially, and would rather describe precisely what we have and have not changed than claim a wholesale redraw. Every caption is now self-contained, stating the quantity plotted, the meaning of any reference line, and the units: Figures 3 and 4 give the no-skill threshold values, NSE of 0 and KGE of -0.41 , and note that persistence is a meaningful reference only over the first two days (1.2); Figure 5 specifies the 2.5th to 97.5th percentile error bars, the pass-inflow assumption, and the compliance band (4.8); Figure 1 states the smoothing window and the SI conversion for cfs (2.17); Figure 6 explains why the lower panel is the operationally relevant case.

On the figures themselves we would welcome more specific guidance. We are willing to change the palette, line styles, text sizes, and output format, and we agree the legibility of axis and legend text at print size could be improved. We have not regenerated them because we did not want to redraw on a guess at the objection and then again after learning what it was.

Reviewer 2

The core work is solid and the practical contribution is genuine. Assuming the major comments are addressed I recommend this manuscript is accepted with minor revision.

Thank you for a detailed and technically careful review. Several of the minor comments identify outright errors in the methods, including one in an equation. Both major comments have changed the manuscript substantially.

2.1 Engagement with the forecast value literature

The stated goal of the framework is to help operators evaluate whether a forecast product will improve operations enough to justify investment. The paper does not include any assessment of forecast value [...] The authors do not need to perform a full value assessment, but they should reference this literature, discuss how it connects to their framework, and provide operators with at least a conceptual pathway from the metrics presented to the investment decision the paper is motivated by.

This is a fair criticism and identifies a genuine gap between what the introduction promises and what the paper delivers. We have added a discussion subsection, “From forecast quality to forecast value,” which draws the distinction (quality is a statistical property of a forecast paired with observations; value depends additionally on human decisions and the cost of errors, so a more accurate forecast can carry less value if the added accuracy falls where the operator would not act differently), cites Farkas et al. (2025) for the economic value literature, and gives operators a two-stage pathway: the real-time quality evaluation establishes which products are plausibly worth adopting and produces the forecast archive needed to drive a subsequent value assessment.

The subsection notes that the tailored metrics are already a step toward value, since forebay exceedance has a direct financial consequence under the FERC license, while remaining a partial valuation because it is not expressed in generation revenue.

We drafted a fuller treatment of specific value metrics, including the relative utility value approach, but judged that a paragraph comparing value metrics we do not apply would read as padding in a paper about quality evaluation; we are glad to restore it if you or the editor prefer. We have not

attempted a value assessment, which would require the utility’s damage function and a model of operator response, returning to the scheduling model coupling under comment 4.9.

2.2 Positioning of the contribution

The five-step iterative process [...] reads more as standard good operational practice than a novel framework contribution. I feel that the tailored metrics and operator feedback loop are where the genuine contribution lies, especially in the real-time context, and these deserve more prominence and deeper treatment.

We agree, and this diagnosis matches Reviewer 3’s and the handling editor’s. Yours says not only what is missing but where the contribution actually is, and we have restructured accordingly. Please see our response to the editor for the new statement of contribution.

The specific changes: steps 1 through 3 are labeled as generic practice for which we claim no novelty. Soliciting feedback and acting on it are separated into “Gather feedback” and “Update metrics,” since collecting operator comment and changing the metric set are different activities and the second is the one that matters, so the process is now six steps. The mechanics of both are documented rather than asserted, including a table tracing each tailored metric to the operator concern that prompted it (3.2), and the two timescales conflated in the iteration step are separated (3.3). We have added the case where a tailored metric separates two products that the generic metrics do not (3.5), a discussion subsection comparing the framework against retrospective evaluation (3.4), and the point that operational reliability is itself part of what a real-time evaluation measures.

2.3 Hyphenation of “real time”

(L2, Title) “real time” should (perhaps) be hyphenated as “real-time” when used as an adjective (also applies throughout the manuscript).

Corrected throughout, including the title, which is now “A real-time reservoir inflow forecast evaluation framework.” Reviewer 4 raised the same point in comment 4.5. The manuscript contained 29 instances of “real time” and none hyphenated; we reviewed each and hyphenated in attributive position, leaving the adverbial phrase (“used in real time”) unhyphenated.

2.4 Verification improving alongside forecasts

(L12–13) The sentences seem to imply that we should expect verification methods to improve alongside improvements in forecasts, but this isn’t the case. [...] Suggest restructuring so the research gap is clearer.

You are right on both counts and the implication was not intended. The passage has been rewritten so the gap is the subject: forecast quality and scheduling have improved while verification practice has not; traditional verification requires hindcasts over a climatologically representative period, often 30 years or more, and their cost and data requirements are why full verifications are not done in practice; the fallback of evaluating archived operational forecasts fits poorly with products that are updated continuously, trained on evolving data streams, hand adjusted by operators, or developed by third parties. The paragraph now closes on the gap itself, which is the absence of a procedure by which operators can evaluate forecasts on their own system under familiar operating conditions.

The point about products delivered as services rather than inspectable models is now supported later in the paper, where we report that neither vendor released model internals or calibration diagnostics.

2.5 Dismissal of ensemble metrics

(L42–44) [...] the dismissal of ensemble evaluation metrics seems premature given that the authors later suggest probabilistic forecasts may help (L170). Consider briefly noting what ensemble metrics could be added if ensemble forecasts were available.

Agreed, and the inconsistency is a fair catch: we set probabilistic evaluation aside early and then invoked probabilistic forecasts as a remedy later. Reviewer 4 asked for the same addition in comment 4.4, so we have added a methods subsection covering CRPS, rank histograms, and reliability diagrams; please see 4.4 for the text.

We have also softened the framing so that the focus on deterministic forecasts is presented as a property of the products under evaluation rather than a judgment about ensemble metrics.

2.6 The argument against hindcast evaluation

(L45–47) The argument against hindcast evaluation is reasonable but somewhat overstated. Hindcast evaluations do not necessarily require operators to re-operate the system, especially if verifying forecasts of environmental variables such as inflows, which you are.

You are correct and the claim as written was overstated. Verifying an inflow forecast against observed inflow does not require re-operating the system; the re-operation requirement applies to the value assessment rather than the quality assessment, which is the distinction your first major comment prompted us to draw. The revised passage says so, and confines the complications to two: where a forecast is modified by human judgment, a hindcast requires forecasters to reconstruct decisions made under past conditions including electricity prices and weather; and extending the evaluation from quality to value requires a reservoir operations model or manual operator input.

A further dependency is worth stating, which we had not made explicit. The calculated inflows we verify against are derived from forebay elevation changes and dam outflow, so the observation record is itself a function of how the system was operated. This does not require re-operation to interpret, but it does mean our reference data is not independent of operations in the way a gauge record would be. We have added this to the limitations.

2.7 Persistence as “arguably worst possible forecast”

(L71) Describing persistence as “arguably worst possible forecast” is incorrect. [...] Suggest rephrasing to “a simple but competitive benchmark.”

Corrected using your suggested wording; see comment 1.2 for the revised passage.

Your point turns out to matter more than a wording fix. Persistence achieves day-ahead NSE of 0.80 at Vernon, competitive with the commercial products, and that is precisely what reveals that forecast A adds almost nothing there once measured against it (comment 1.6). Treating persistence as a floor rather than a real competitor is what had hidden the result.

2.8 Smoothing window width

(L80–81) A triangular smoother is reasonable but the choice of window width is not discussed. [...] Please note the window used and perhaps the sensitivity to this choice on your findings.

Reviewer 4 raised this as their first major comment and we have addressed it with both the value and a sensitivity analysis. The window is 24 hours, implemented as two successive 24-hour moving averages of the hourly calculated inflow, which produces the triangular weighting, and was chosen to match the day-ahead operational timescale. Metrics improve modestly with window width, the largest change being 0.05 NSE, and the ranking of the three products is unchanged at every width

at every location. Please see comment 4.1 for the table.

2.9 KGE name and citation

(L92) “King-Gupta Efficiency” should be “Kling-Gupta Efficiency.” Also, “(refs)” is a placeholder that needs to be replaced with actual references.

Both corrected, with Gupta et al. (2009) as you surmised, and Nash and Sutcliffe (1970) added for NSE in the same sentence. Reviewer 4 flagged the same two items in comment 4.6.

2.10 The claim about KGE and large values

(L92–93) The claim that KGE is not skewed by large values while NSE and RMSE are requires clarification. KGE and NSE both use correlation as a component. The relevant difference is that KGE decomposes performance into correlation, variability ratio, and bias ratio rather than using squared error. The statement as written is misleading.

You are right, and your characterization is the correct one. The original sentence attributed the difference to the squared error term, but KGE also incorporates correlation, which is itself computed from squared deviations, so the stated reason does not hold. We have replaced it with your framing:

“We recommend KGE because it decomposes forecast performance into correlation, a variability ratio, and a bias ratio, so that a low score can be attributed to a specific deficiency rather than only registered as large aggregate error. NSE and RMSE aggregate squared error into a single number, which makes them more strongly influenced by the largest errors and less diagnostic about their source.”

We have also added a sentence making clear that KGE and the NSE-based skill score answer different questions and that we recommend reporting both: KGE to diagnose where a forecast is deficient, and a skill score against an explicit benchmark to compare products and locations (comment 1.6).

2.11 Placement of the Knoben et al. citation

(L95–96) The KGE threshold of -0.41 is attributed to comparing against the mean of observations but the Knoben et al. (2019) citation is on L97, not adjacent to the claim.

Corrected. The citation now sits with the threshold statement it supports, followed by the consequence, previously left implicit: because the threshold is not zero, NSE and KGE should not be used interchangeably and intuition built on NSE values does not transfer to KGE. Knoben et al. (2019) also bears directly on the cross-location comparability point raised in comment 1.6, so we cite it in the benchmark section as well.

2.12 Bias versus error

(L98–99) “The bias of a single forecast point is the difference between observed and forecasted values” defines error (sometimes called residual), not bias.

Corrected. The text now reads:

“The error, or residual, of a single forecast is the difference between the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts as a percentage of total observed volume.”

The original definition also had the sign backwards relative to the PBIAS equation that follows it.

The revised wording is consistent with the equation.

2.13 Pearson correlation formula

(Eq. 3) The Pearson correlation coefficient formula is missing a summation over the second term of denominator.

Corrected, and thank you for checking the equations. The denominator was written as a single summation over the product of squared deviations, and now reads as the product of two separate sums:

$$r = \frac{\sum(o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum(o_i - \mu_o)^2 \sum(f_i - \mu_f)^2}}$$

We have checked the remaining equations; the KGE and PBIAS expressions are correct as published. The reported correlation values were computed with a standard library function and are unaffected by the typesetting error.

2.14 Nominal capacity

(L107) 589 megawatts of “nominal” hydropower capacity. Is this installed capacity or average generation?

It is installed nameplate capacity. “Nominal” was ambiguous and has been replaced with “589 megawatts of installed nameplate hydropower capacity across 13 generating stations,” which also makes clear that the system total covers many more facilities than the three evaluated here.

2.15 Relationship among the three sites

(L109) Add a sentence describing how these 3 sites are related [...] Also good to note why you chose these sites over all the other sites (green dots in the figure).

Agreed, and this omission made the results harder to interpret than necessary. Your inference is correct: the three dams lie in series on the Connecticut River mainstem, Wilder upstream, then Bellows Falls, then Vernon, and are operated jointly, with releases from each becoming inflow to the next. This is why the drainage areas are nested and why bias propagates downstream, an effect we noted in the results without giving the reader the information needed to understand it. We now quantify the propagation (1.5).

On site selection, these three are the largest generators in the system, they are the dams subject to the FERC forebay constraint that motivated the tailored metrics, and they are where the operators concentrate their scheduling attention. The three tributaries were included because they are the only significant local inflows, entering between Wilder and Bellows Falls. The remaining facilities are smaller and not covered by the same license conditions. Both points are now in the text.

2.16 Anonymization and reproducibility

(L117) Anonymising the commercial forecast products is understandable but limits reproducibility. Consider at minimum describing the general class of model used by each [...] so readers can contextualise the results.

Agreed. Reviewers 1 and 4 asked for the same thing; please see comment 1.4 for the text we have added and for what we cannot disclose. We have also added the inability to inspect commercial products to the limitations, since it constrains any evaluation of this kind and is part of what motivates judging products on operational performance.

2.17 SI units

(Figure 1) Y-axis units are “cfs” (cubic feet per second). Consider also providing SI units for international readers.

Added. The Figure 1 caption gives the conversion ($1,000 \text{ cfs} \approx 28.3 \text{ m}^3/\text{s}$), and SI equivalents appear at first use in the text, including the 0.5 foot forebay limit as approximately 15 cm and the largest observed inflow in both units. We have kept cfs as the primary unit because it is what the operators and the FERC license use, and the tailored metrics are in feet of forebay for the same reason. We have not added a secondary axis to the figure itself; if the editor prefers that we will do so.

2.18 Seasonal performance

(L134) “Forecasts tend to perform worse in the winter which is the wet season and worse in the summer which is the dry season.” This says forecasts perform worse in both wet and dry seasons. Please clarify when forecasts actually perform well and why.

You are right that the sentence contradicts itself and the logic was garbled. Re-examining the monthly results showed the original characterization of the seasons was also wrong: winter is not the wet season in this basin. Mean inflow at Vernon is highest in March and April and lowest from September through November, so the high-flow season is spring, driven by snowmelt, and that is where skill is highest for all three products.

The rewritten passage states this, and explains the mechanism: performance is poorest in the seasons where inflow is driven by precipitation events, rain and rain-on-snow in winter and convective storms in summer, both of which depend on highly uncertain precipitation forecasts, whereas snowmelt-driven inflow is comparatively predictable because it is governed by an accumulated snowpack observable in advance that melts over weeks, so skill depends more on temperature, which is forecast more reliably.

Two further corrections came out of this. The original text gave the snowmelt season as April through July, which does not match the monthly inflow pattern; it is now March through May. And the January and February KGE values for forecast A were given as 0.39 and -0.40 ; recomputing gives 0.45 and -0.67 . The revised value is also consistent with the figure, in which the February point falls below the plotted range, which -0.40 would not.

2.19 The inflow-equals-outflow assumption

(L146) [...] the forebay exceedance analysis assumes an inflow=outflow condition which is operationally unrealistic as the authors note. It would strengthen the analysis to discuss how operators actually deviate from this assumption and what the real-world exceedance rates might be.

A fair point. Operators do not hold pass-inflow conditions: they schedule generation to a market bid, which means departing from inflow during the day and recovering the forebay overnight, and they intervene when the forebay approaches the band. The text now states that the pass-inflow assumption is not how the system is operated, that the metric is idealized and intended as an upper limit on the compliance risk a forecast carries rather than a prediction of observed exceedance rates, and that capturing actual risk requires a scheduling model with direct operator oversight.

We consider this the right quantity for comparing products because it isolates forecast quality from the operators’ skill at correcting for it. Realized exceedance is less frequent than these probabilities imply, but realized rates reflect both the forecast and the effort spent compensating for it, and that

effort is itself a cost of a poorer forecast. We have not estimated real-world rates, which would require representing intervention behavior and so returns to comment 4.9.

2.20 Extremes and the record length

(L178–182) [...] Consider also noting the limitation that almost two years of data may not capture the full range of extreme events which are often the conditions where forecast quality matters most for operational decisions.

Agreed, and we have stated the limitation explicitly. Reviewer 4 asked in comment 4.2 what the period did contain, so we addressed both together; please see that response for the events observed and their position in the long-term annual peak record. In short, the record covers a useful range but not the upper tail, and no severe multi-year drought.

One point specific to this system is worth adding: representing multiple high flow events matters less at GRH than it might elsewhere, because operational flexibility is limited during high flows and operators discount forecast errors in those conditions. The gap that does matter for GRH is the absence of a strong multi-year drought, which we state as a limitation.

Reviewer 3

However, proposed framework itself is not well described, such as what is new, different from existing ones, and comparison of past studies are insufficient.

Thank you for this review. Your observation that we implied the framework was novel without describing the novelty was among the most useful comments we received, and the handling editor highlighted it as well. Our response to the editor describes the new statement of contribution. Comments 3.4 and 3.5 have produced what we now consider the central result of the paper.

3.1 “market prices”

(l47) “market prices” — Could you please clarify what it means?

The sentence was too compressed. The point concerns why a hindcast evaluation is difficult for a hydropower operator: reconstructing what a forecaster would have done in a past period requires knowing the conditions they faced, and those include the wholesale electricity prices they were bidding against, which drive the generation schedule and hence release decisions. Those price signals, and the operator’s response to them, are not generally recoverable after the fact. The revised passage is described under comment 2.6, which prompted a further correction to the same sentences: the claim that a hindcast requires re-operating the system was overstated for a quality assessment, and the requirement properly applies to a value assessment.

3.2 How feedback is reflected in the process

(§2.1) Step 1 through 3 are so general. I assume that steps 4 and 5 are unique points in this manuscript. Please provide a more detailed explanation of how feedback is reflected in step 4.

You have identified the structure of the contribution correctly, and we now say so directly rather than presenting all steps as equally novel. Steps 1 through 3 are labeled as generic.

Your comment also exposed an ambiguity: what was a single step for soliciting feedback conflated gathering operator comment with actually changing the metric set. We have split them into “Gather feedback” and “Update metrics,” so the process is now six steps, with 4 through 6 being where it departs from retrospective verification.

Both steps are expanded with the mechanics. Feedback was collected in recurring meetings, monthly for most of the evaluation, each reviewing the current metrics against recent operating experience, with operators asked which recent forecast errors mattered operationally and which did not. Their answers frequently diverged from what the generic metrics indicated, especially in the high-flow season: they discounted large errors during spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to breach the forebay compliance band as serious. That is what motivated the tailored metrics, the exceedance metrics in particular.

We have also added a table to the manuscript tracing each tailored metric to the operator concern that prompted it and the approximate month it was added, so the process is auditable rather than asserted: day-ahead volumetric skill, because market bids are set from next-day volume rather than hourly shape (months 1–2); the seasonal breakdown, because products were suspected to behave differently in snowmelt and rain driven regimes (3–4); cumulative forebay error, because average error does not indicate whether the band would be breached (4–6); and exceedance from a non-neutral start, because operators rarely begin a day at mid-band (5–6).

We should be candid that the months are approximate: the evaluation predates this paper and we kept no formal log of metric provenance, so these are reconstructed from meeting materials and correspondence, though the sequence and motivations are accurate. A practical recommendation follows, which we now state: an evaluation of this kind should log metric provenance as it goes, since the record of why each metric exists is part of what makes the results defensible later.

3.3 Is a one-year learning period required?

(“5. Iterate”) This implies that at least one year learning period is required. Is this understanding correct?

Your understanding is correct for a full evaluation, and your question identified that the original wording conflated two durations. The manuscript now separates them: metric development was complete within roughly the first six months, after which iteration served to accumulate data rather than to identify new metrics, while a minimum defensible skill assessment requires at least one full annual cycle because performance varies substantially by season and products may rank differently in each. Operators should therefore expect the metric set to stabilize well before the evaluation period ends.

Our own results bear on the second point. Pooled day-ahead KGE for forecast A ranges from -0.67 in February to 0.88 in May, so an evaluation conducted in spring alone would have reached a substantially more favorable conclusion about that product than the full record supports. We have also noted the tension the operators raised, that industry preference runs toward shorter evaluations, sometimes a single season, while a single season samples only one hydrologic outcome.

3.4 Comparison with traditional methods

(result of case study) Were all the results derived through the proposed framework? If so, please compare those result with those obtained via “traditional” methods.

All results were produced through the framework, and we agree the comparison you describe is what demonstrates its effectiveness. We have added it in two forms.

First, a discussion subsection, “Comparison with retrospective forecast evaluation.” A retrospective evaluation provides larger samples, a wider range of hydrologic conditions, and better coverage of extremes, since hindcasts can span decades where ours spans less than two years. Against

that, translating thousands of historical forecasts into operational decisions may be prohibitively expensive, and even where historical operations could be reconstructed it is difficult or impossible to reconstruct the exact market conditions at the time of a decision. A real-time evaluation reflects actual operational conditions including outages and market feedbacks, and requires no additional data or operator effort. We are explicit that it is not a replacement for retrospective verification where a hindcast is feasible, but a practical alternative where one is not.

Second, the comparison of conclusions. A conventional verification of these forecasts, computing NSE, KGE, PBIAS, and correlation over the full record, concludes that forecast B outperforms forecast A and is comparable to the in-house forecast. That conclusion is correct and this framework reproduces it. What the conventional presentation omits shows up in two places: the compliance risk finding that carried the most weight with the operators (3.5), and the fact that raw efficiency metrics misidentify where forecast A is weakest, which only the benchmark-relative score reveals (1.6). We do not overstate this: the approaches agree on the overall ranking, and what the framework adds is dimensions of comparison the conventional summary leaves invisible rather than a contradiction of it.

3.5 Advantage of tailored metrics

(§3.1 Tailored metrics) Please explain the advantage of introducing tailored metrics. For example, it would be helpful to show a case where using tailored metrics led to a different forecast conclusion compared to general metrics.

This was the right thing to ask for and we have added that case as a result rather than leaving it for the reader to infer. To be precise about the claim: the generic and tailored metrics do not reach opposite conclusions about which product is better, since forecast B outperforms forecast A on both. What the tailored metric does is separate the products in a dimension the generic metrics leave invisible, in units the operators are accountable for.

The forebay exceedance metric is that case. From a neutral forebay starting position, the probability of breaching the compliance band is below 10% for every product at every lead time in the 1–3 day range, and on that basis the products are difficult to distinguish. From the non-neutral position operators are routinely in, they separate sharply: at Bellows Falls forecast A reaches 23.7% on day 1 and 11.9% on day 2, against 1.8% and 2.4% for forecast B; at Vernon it reaches 10.1% by day 3 against 5.2%. A roughly tenfold difference in compliance risk is invisible in the NSE, KGE, PBIAS, and correlation values, none of which know that a penalty attaches to a particular forebay threshold.

That non-neutral case exists only because the operators asked for it; the neutral-start case, which is what an analyst would naturally compute and what our original draft emphasized, shows nothing. This is the mechanism by which operator participation in metric design changes the result rather than merely improving its reception.

On comparison to past studies, the introduction already discussed the head-to-head evaluation exercises (the HEPEX community experiments and the Bureau of Reclamation forecast rodeos), and we have sharpened what distinguishes this work: those evaluate forecast quality in general, while this framework evaluates fitness for one utility’s operating constraints under live conditions.

Reviewer 4

The overall idea is relevant and practically useful for hydropower operators seeking to assess competing forecast products. The paper is generally well-written, and the iterative, operator-

inclusive evaluation approach adds value.

Thank you for the careful reading and for the specific, actionable comments. We have addressed all nine. Comment 4.2 led us to check the evaluation period against the long-term record, which corrected a claim we would otherwise have overstated.

4.1 Smoothing window size and sensitivity

Lines 80–81 recommend a triangular smoother for calculated inflows but do not specify the smoothing window size [...] The window size must be stated explicitly, and a sensitivity analysis or justification for the chosen value would strengthen the paper.

You are right that this parameter was left unstated and that it affects the metrics. We have added both the value and a sensitivity analysis. Reviewer 2 raised the same point in comment 2.8.

The smoothing is a 24-hour triangular window on hourly calculated inflow, implemented as two successive 24-hour moving averages. The window was matched to the operational timescale: the day-ahead volumetric forecast is the quantity of interest, so smoothing at 24 hours removes sub-daily forebay noise without attenuating the daily signal being evaluated. Day-ahead NSE at each dam over four window widths:

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

Metrics improve modestly with window width, as expected since smoothing removes observation noise that no forecast can reproduce; the largest change is 0.05 NSE (Bellows Falls, forecast A). The improvement is not strictly monotonic, the in-house forecast at Bellows Falls dipping from 0.84 to 0.83 between 0 and 6 hours before recovering, which we attribute to sampling rather than to any property of the smoother. The ranking is unchanged at every width at every location, and the gap between the commercial products is stable: forecast B leads A by 0.39 NSE at Bellows Falls with no smoothing and by 0.37 with 24-hour smoothing. The 24-hour column reproduces Table 1 exactly, so the sensitivity analysis and the main results come from the same pipeline.

We have added the table and the following caution, which is the more generally useful part of the answer:

“Evaluators should nonetheless state the window explicitly and check that their conclusions are insensitive to it, since a window wide enough to attenuate the signal of interest would flatter forecasts that under-predict variability. We recommend matching the window to the decision timescale rather than choosing the window that maximizes metric values.”

4.2 Hydrologic representativeness of the evaluation period

Line 116 states the evaluation ran from July 2023 to May 2025. The authors should discuss whether this period captured any notable extreme hydrologic events [...] and how climato-

logically representative it is relative to the long-term record.

This was a valuable comment and answering it properly changed two things in the manuscript.

First, the stated period was wrong. The record analyzed runs from June 2023 to April 2025, and the three products do not cover it uniformly: forecast B spans the full period (657 day-ahead forecasts per dam) while forecast A and the in-house forecast both end in November 2024 (539 and 528 respectively). We have corrected the period, added the per-product sample sizes to Table 1, and verified that restricting the comparison to the 510 days on which all three products are present leaves the ranking unchanged at all three dams, with individual metrics shifting by no more than 0.03 NSE. We now report this partial coverage as a characteristic feature of real-time evaluation rather than smoothing over it.

Second, on the events. A new subsection reports that the period captured the July 2023 Vermont flood, which produced the largest calculated inflows of the evaluation at Vernon (83,100 cfs, 2,353 m³/s) and Bellows Falls (67,000 cfs), and a December 2023 rain-on-snow event that produced the largest inflow at Wilder (27,600 cfs), along with spring freshets in 2024 and 2025 and dry summers in 2023 and 2024. Checked against the long-term annual peak series at the USGS reference gages, the July 2023 event ranks among the ten largest flows on record on the main stem, though below the 1936, 1938, and 1927 floods and the 2011 Tropical Storm Irene peak. The period was therefore not hydrologically quiet, but neither did it sample the upper tail, and the manuscript now states that the conclusions are conditional on the observed period. Reviewer 2 asked in comment 2.20 for the limitation to be stated explicitly, which we have added to the limitations discussion.

One caveat: the gage annual peak series are instantaneous peak discharge at the gage, while our values are 24-hour smoothed calculated inflow at the dams, so the two are not strictly commensurable. We have therefore phrased the manuscript statement as an approximate ranking rather than a numerical comparison.

4.3 Characterization of the commercial forecasts

Line 117 [...] the paper should provide at least a general characterization of each forecast’s methodology (e.g., physics-based, machine-learning-based, hybrid, statistical).

Agreed, and Reviewers 1 and 2 raised the same point. Please see comment 1.4 for the characterization we have added and for what we cannot disclose. In summary: forecast A is a conceptual rainfall-runoff model with hydraulic routing driven by commercial numerical weather prediction, issued once daily out to roughly 8 days; forecast B is a machine learning model trained on gauge and remotely sensed observations, issued twice daily out to 10 days; the in-house forecast combines RFC forecasts with routing and operator judgment and extends about 32 hours. The issue frequencies and horizons are taken from the forecast archive rather than vendor documentation. We have also added that neither vendor released model internals or calibration diagnostics, which is relevant to the framework’s motivation rather than merely a caveat.

4.4 Extension to probabilistic and ensemble forecasts

The paper focuses only on deterministic forecast evaluation [...] the framework could be briefly extended to probabilistic or ensemble forecasts using metrics such as CRPS, reliability diagrams, or rank histograms.

We agree and have added a methods subsection, “Extension to probabilistic and ensemble forecasts.” Reviewer 2 asked for the same in comment 2.5. It makes the point that the framework is agnostic to whether the forecast is deterministic or probabilistic, since only the metrics and

visualizations change and not the process, and notes that CRPS is the natural accuracy measure for ensembles, with the useful property that it reduces to mean absolute error for a deterministic forecast so the two can be compared on a common scale, alongside rank histograms and reliability diagrams for assessing ensemble properties.

Two connections are worth drawing out. The forebay exceedance metric we develop is already a probabilistic statement derived from a deterministic forecast, computed across the empirical distribution of that forecast’s past errors; given an ensemble it would be computed directly from the members, which is both simpler and better founded. And operators bidding into a day-ahead market must submit a single quantity, so a probabilistic forecast must be reduced to a point value at the moment of the bid, which is why improvements in deterministic skill remain directly useful. This is a description of how the framework would extend rather than a demonstration, since all three products evaluated here are deterministic.

4.5 Hyphenation of “real time”

Throughout the manuscript: “real time” should be hyphenated when used as a compound adjective. Usage is inconsistent.

Corrected throughout, including the title. Reviewer 2 raised the same point in comment 2.3, where we describe the treatment.

4.6 KGE citation

(Methods, line 92) Remove the placeholder “(refs)” and insert the actual KGE citation (Gupta et al. 2009).

Corrected. The placeholder is replaced with Gupta et al. (2009), and we have added Nash and Sutcliffe (1970) for NSE in the same sentence. We also corrected the misspelling Reviewer 2 caught in comment 2.9: “King-Gupta Efficiency” now reads “Kling-Gupta Efficiency.”

While checking this we found and fixed a broken citation key elsewhere: the introduction cited `kratzertLearningUniversalRegional2019` where the bibliography entry is `kratzertLearningUniversalRegional` so that citation was not resolving. All citation keys now resolve against the bibliography.

4.7 RMSE or MAE in Table 1

(Table 1) Consider adding RMSE or MAE values alongside the existing metrics to provide a sense of forecast error magnitude in physical units.

Agreed, and this was a real omission since every metric in the original table was dimensionless. We have added RMSE in cfs to Table 1. The values are informative in themselves: forecast A’s day-ahead RMSE at Bellows Falls is 5,904 cfs against 2,615 for forecast B, which conveys the practical size of the difference more directly than the NSE values of 0.55 and 0.91.

The RMSE column turned out to serve a second purpose. Because the NSE skill score added under comment 1.6 is a ratio of mean squared errors, a reader can recover that column from the RMSE column as $1 - (\text{RMSE}_f / \text{RMSE}_{ref})^2$, which makes the new column checkable from the table itself. Table 1 also gains the per-product sample size described under comment 4.2.

4.8 Figure 5 caption

(Figure 5 caption) The caption does not specify that the error bars represent the 2.5th and 97.5th percentiles. Figure caption needs to be self-contained.

Corrected. The Figure 5 caption now states that points show the median across all forecasts and

error bars span the 2.5th to 97.5th percentile, and specifies the pass-inflow assumption and the ± 0.5 foot compliance band. We reviewed all captions for self-containment as part of the caption revision described under comment 1.8.

4.9 What coupling to scheduling models would entail

(Discussion) The authors note that coupling this evaluation with hydropower scheduling models is an important next step. This point could be strengthened by briefly discussing what such a coupling would entail and what data would be needed.

Agreed, the original text asserted the next step without describing it. We have expanded the passage:

“One difficulty in this approach is that scheduling models may be configured to reproduce a utility’s current bidding strategy, which was itself developed around the accuracy of its existing forecast. A more accurate forecast may show little revenue benefit simply because the model is not set up to exploit it. Quantifying the value of improved forecasts may therefore require revising the scheduling model and the operating strategy alongside any new forecast.”

On the data required, such a coupling would route each candidate forecast through a scheduling model to produce the generation schedule it implies, then value that schedule against realized inflows and market prices, so that forecast differences are expressed in revenue rather than skill. This needs the utility’s scheduling model with its operating constraints and unit efficiency curves, historical day-ahead and real-time market prices at the relevant node, the realized inflow record, and the archive of forecasts as issued so that schedules use only information available at bid time. That last requirement is a further reason a real-time evaluation is a useful precursor, since it produces exactly this archive as a byproduct. The same coupling is what a full value assessment (2.1) and an estimate of realized forebay exceedance rates (2.19) would both require.

Summary of changes

Framing. Statement of contribution added to the introduction and the gap statement rewritten (editor, 1.1, 2.4). Process restructured from five steps to six, separating “Gather feedback” from “Update metrics,” with steps 1–3 labeled generic and novelty claimed only for 4–6 (2.2, 3.2). New discussion subsections comparing the framework with retrospective evaluation (3.4) and connecting quality to value, citing Farkas et al. (2025) (2.1). Abstract revised.

New analysis. NSE skill score against persistence added to Table 1 with a defining equation; it inverts the apparent cross-site ordering of forecast A, strongest at Wilder (0.85) and adding almost nothing at Vernon (0.03) where its raw NSE is highest (1.6). Forebay exceedance reported as a result that separates the two commercial products where generic metrics do not (3.5). Smoothing sensitivity table over 0, 6, 12, and 24-hour windows (4.1, 2.8). RMSE in cfs, per-product sample sizes, and a common-period check added (4.7, 4.2). Downstream bias propagation quantified (1.5). New subsection on the hydrology of the evaluation period (4.2, 2.20).

New descriptive material. New subsections on operating context (1.3), forecast products (1.4, 2.16, 4.3), and the extension to ensemble forecasts (4.4, 2.5). New tables of basin characteristics (1.5) and tailored metric provenance (3.2). Expanded treatment of feedback mechanics (3.2), iteration timescales (3.3), the scheduling model coupling (4.9), and the pass-inflow assumption (2.19).

Corrections. Evaluation period corrected to June 2023–April 2025 (4.2). NSE values mislabeled as KGE in the cross-site paragraph, now rewritten (1.6). Monthly KGE for forecast A in January and February corrected to 0.45 and -0.67 , and the snowmelt season to March–May (2.18). Pearson correlation denominator corrected (2.13). Bias definition corrected and made sign-consistent with PBIAS (2.12). KGE rationale corrected (2.10). “King-Gupta” $\rightarrow$ “Kling-Gupta,” “(refs)” replaced with Gupta et al. (2009) and Nash and Sutcliffe (1970), and a broken citation key fixed (2.9, 4.6). Knoben et al. (2019) moved adjacent to the threshold it supports (2.11). Persistence redescribed (2.7, 1.2). Capacity clarified as installed nameplate (2.14). Site relationships and selection explained (2.15). “real time” hyphenated attributively throughout (2.3, 4.5). SI units added (2.17). Figure captions made self-contained (1.8, 4.8). Tables rebuilt (1.7).

Requests we were not able to fully satisfy

We would rather name these than let them pass as addressed.

- **Climatological benchmark in Figures 3 and 4** (1.2). Recommended in the text but not added: a defensible daily climatology of calculated inflow at the dams needs a record that begins only with the evaluation, and a gage-derived substitute would not be commensurable.
- **Calibration and validation diagnostics for the commercial products** (1.4). Not released to us by either vendor.
- **Redrawing the figures** (1.8). Captions substantially revised, figures not regenerated; we would welcome specific guidance on which aspects are considered deficient.
- **A full forecast value assessment** (2.1). Literature cited and a two-stage pathway given, but no value metric computed.
- **Realized forebay exceedance rates** (2.19). Needs the scheduling model coupling of 4.9.
- **Demonstrated probabilistic evaluation** (2.5, 4.4). Described but not demonstrated, as all three products evaluated are deterministic.

We thank the reviewers again for the care they took. In four places their questions caused us to find and correct errors that would otherwise have gone to print: the stated evaluation period, the seasonal characterization and its monthly values, the mislabeling of NSE as KGE in the cross-site comparison, and the Pearson correlation equation. Reviewer 1’s request that we check the NSE values (comment 1.6) was the most productive comment we received. Following it up produced the skill score column, corrected the paragraph that discussed those values, and led to what we now regard as one of the paper’s more useful practical findings: a commercial product can post its highest efficiency score at precisely the location where it adds least to what the operator already has.